



# Why Sine and Cosine?

## Fourier's Core Insight (1822)

**Every** signal — no matter how complex — can be expressed as a weighted sum of sine and cosine waves of different frequencies.

- Think of it like a **recipe**: the signal is the dish, and sines/cosines are the ingredients.
- A **prism** splits white light into colours; the Fourier Transform splits a signal into its frequency components.
- **Any real-valued signal** needs only real cosines (even part) and real sines (odd part).

## Musical Analogy

A guitar chord contains many individual notes (frequencies) played at once. Fourier analysis *separates* them — telling you exactly which notes are present and how loud each is.

# The Sine Wave

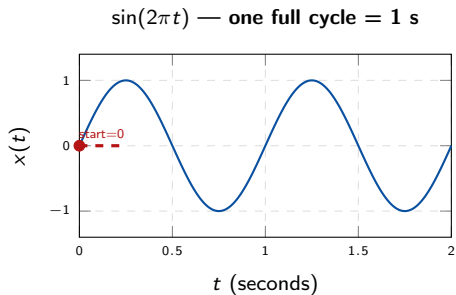
## Formula

$$x(t) = A \sin(2\pi f t + \phi)$$

- $A$  Amplitude (peak height)
- $f$  Frequency (Hz, cycles/s)
- $t$  Time (seconds)
- $\phi$  Phase offset (radians)

## Key Properties

- **Starts at 0**, rises first
- **Odd function:**  $\sin(-t) = -\sin(t)$
- Represents the **imaginary** part in complex notation



# The Cosine Wave

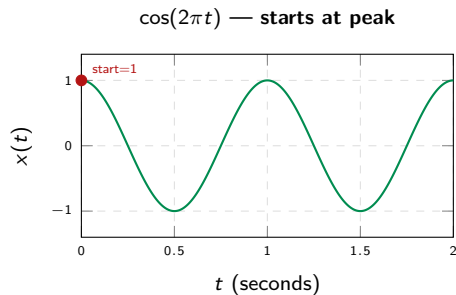
## Formula

$$x(t) = A \cos(2\pi f t + \phi)$$

- $A$  Amplitude
- $f$  Frequency (Hz)
- $t$  Time
- $\phi$  Phase offset

## Key Properties

- **Starts at maximum** value  $A$
- **Even function:**  $\cos(-t) = \cos(t)$
- Represents the **real** part in complex notation
- Just a **phase-shifted** sine:  $\cos(t) = \sin(t + \frac{\pi}{2})$



# Sine + Cosine = Any Signal!

## General Fourier Synthesis Formula

$$x(t) = a_0 + \sum_{n=1}^N \left[ \underbrace{a_n \cos(2\pi n f_0 t)}_{\text{even / symmetric}} + \underbrace{b_n \sin(2\pi n f_0 t)}_{\text{odd / anti-symmetric}} \right]$$

### Parameters:

- $a_0$  — DC offset (average value)
- $a_n$  — cosine weights
- $b_n$  — sine weights
- $f_0$  — fundamental frequency
- More terms  $\rightarrow$  better reconstruction

### Square Wave Example

A square wave decomposes into:

$$\frac{4}{\pi} \left[ \sin t + \frac{\sin 3t}{3} + \frac{\sin 5t}{5} + \dots \right]$$

Only *odd* harmonics, no cosines!

# Two Independent Signal Properties

Every signal can be described along **two independent axes**:

## Axis 1 — Time Sampling

### Continuous

Defined at every instant of time.

→ Analog signals (e.g., vinyl record)

### Discrete

Defined only at integer sample indices.

→ Digital signals (e.g., MP3 file)

## Axis 2 — Repetition

### Periodic

Repeats with period  $T$ :  $x(t) = x(t + T)$

→ Infinite repetition

### Aperiodic

Does *not* repeat.

→ Finite burst or non-repeating

$$\underbrace{\text{Continuous or Discrete}}_{\text{time axis}} \times \underbrace{\text{Periodic or Aperiodic}}_{\text{repetition}} = \boxed{4 \text{ combinations}}$$

# Signal Type 1 — Aperiodic & Continuous

## Definition

Defined for *all*  $t \in (-\infty, +\infty)$ . **Does not** repeat — no period  $T$ .

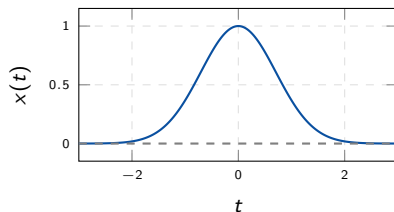
- Smooth, unbroken curve
- Exists for a finite duration *or* extends forever without repeating
- Examples: radar pulse, decaying exponential, Gaussian bell curve
- Spectrum is **continuous and aperiodic**

## Transform Used

**Fourier Transform (FT)**

$$X(f) = \int_{-\infty}^{+\infty} x(t) e^{-j2\pi ft} dt$$

**Gaussian Pulse (aperiodic, continuous)**



# Signal Type 2 — Periodic & Continuous

## Definition

Smooth, unbroken curve that **repeats** with a fixed period  $T$ .

$$x(t) = x(t + T) \quad \forall t$$

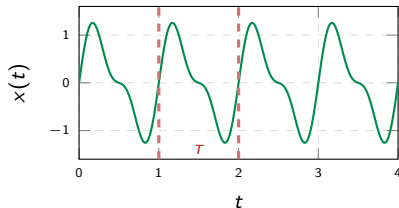
- Extends infinitely in both time directions
- The pattern looks exactly the same in each period
- Examples: pure sine wave, square wave, sawtooth wave, AC mains voltage
- Spectrum is **discrete and aperiodic** (only multiples of  $f_0 = 1/T$ )

## Transform Used

Fourier Series (FS)

$$c_n = \frac{1}{T} \int_0^T x(t) e^{-j2\pi nt/T} dt$$

Repeating Wave (periodic, continuous)





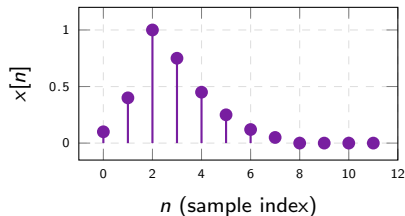
# Signal Type 3 — Aperiodic & Discrete

## Definition

Defined only at **integer sample indices**  $n$ . **Does not repeat.**

- Written  $x[n]$  — note the **square brackets**
- A sequence of individual numbers, not a curve
- Examples: digitised single pulse, unit impulse  $\delta[n]$ , FIR
- Spectrum is **continuous and periodic** (period  $2\pi$  in normalised frequency)

Discrete Aperiodic Signal



## Transform Used

DTFT

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

# Signal Type 4 — Periodic & Discrete

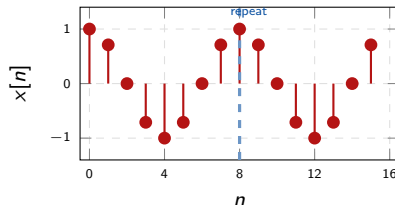
## Definition

Defined at **integer sample indices** and **repeats** with period  $N$  samples.

$$x[n] = x[n + N] \quad \forall n$$

- $N$  = number of samples per period
- This is **exactly what computers store and process**
- The DFT *assumes* the  $N$ -sample block repeats
- Spectrum is also **discrete and periodic** — it has exactly  $N$  frequency points

Discrete Periodic Signal ( $N = 8$ )



## Transform Used

Discrete Fourier Transform (DFT)

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

# The Four Fourier Transforms — Overview Table

| Name                          | Time Domain               | Freq. Domain              | Application                 |
|-------------------------------|---------------------------|---------------------------|-----------------------------|
| <b>Fourier Transform (FT)</b> | Continuous Aperiodic      | Continuous Aperiodic      | Analog signal analysis      |
| <b>Fourier Series (FS)</b>    | Continuous Periodic       | <b>Discrete</b> Aperiodic | Periodic analog signals     |
| <b>DTFT</b>                   | <b>Discrete</b> Aperiodic | Continuous Periodic       | Infinite digital sequences  |
| <b>DFT</b>                    | <b>Discrete Periodic</b>  | <b>Discrete Periodic</b>  | <b>Computers &amp; DSP!</b> |

Golden Rule (memorise this!)

**Discrete** in one domain  $\iff$  **Periodic** in the other domain.

**Continuous** in one domain  $\iff$  **Aperiodic** in the other domain.